Eigen Space & Characteristic Equation

Introduction

The eigenspace is the space generated by the eigenvectors corresponding to the same eigenvalue - that is, the space of all vectors that can be written as linear combination of those eigenvectors

Eigenspace is a fundamental concept in linear algebra that arises in the context of eigenvalues and eigenvectors of a matrix. These concepts are crucial in various fields such as physics, engineering, computer science, and data analysis, particularly in the study of linear transformations and systems of linear equations.

Introduction

Eigenspace, also known as the eigen subspace, is the set of all eigenvectors associated with a particular eigenvalue, along with the zero vector. In other words, it is a vector space formed by eigenvectors corresponding to the same eigenvalue and the origin point. The eigenspace corresponding to the eigenvalue λ of the matrix A can be found by solving the equation $(A - \lambda I)v = 0$, where I is the identity matrix of the same dimension as A and v is the vector. The solutions to this equation form the eigenspace of λ.

Introduction

Properties of Eigenspace

The eigenspace has several important properties:

- **Dimensionality:** The dimension of the eigenspace is called the geometric multiplicity of the eigenvalue. It is determined by the number of linearly independent eigenvectors associated with that eigenvalue.

- **Subspace:** An eigenspace is a subspace of the vector space on which the matrix A acts. This means that it is closed under vector addition and scalar multiplication.

- **Null Space:** The eigenspace corresponding to the eigenvalue λ is essentially the null space (kernel) of the matrix $(A - \lambda I)$.

Introduction

Applications of Eigenspace

Eigenspaces are used in various applications, including:

•**Stability Analysis:** In systems theory, eigenvectors and eigenvalues are used to analyze the stability of equilibrium points.

•**Principal Component Analysis (PCA):** In statistics and machine learning, PCA uses eigenvectors and eigenvalues to perform dimensionality reduction for data visualization and noise reduction.

•**Quantum Mechanics:** In quantum mechanics, eigenvectors represent possible states of a quantum system, and eigenvalues correspond to observable quantities like energy.

•**Vibration Analysis:** In engineering, eigenvectors and eigenvalues are used to determine the modes of vibration of structures.

Introduction

Computing Eigenspace

To compute the eigenspace of a matrix, one must first determine its eigenvalues by solving the characteristic equation $\det(A - \lambda I) = 0$. Once the eigenvalues are found, the eigenspace for each eigenvalue is computed by solving the system of linear equations $(A - \lambda I)v = 0$ for the vector *v*. The solutions to these systems form the basis for the eigenspaces.

Eigenspace

Suppose that A is a square matrix. We already know how to check if a given vector is an eigenvector of A and in that case to find the eigenvalue. Our next goal is to check if a given real number is an eigenvalue of A and in that case to find all of the corresponding eigenvectors.

Square Matrix (A)	$\Longrightarrow$	Check if a vector is Eigenvector (ν)	$\Longrightarrow$	Find the Eigenvalue (λ)

Given a real number & a Square Matrix (A)	$\Longrightarrow$	Check if it is an Eigenvalue (λ) of A	$\Longrightarrow$	Find the corresponding Eigenvector (ν)

Eigenspace

Let A be an $n \times n$ matrix, and let λ be a scalar. The eigenvectors with eigenvalue λ, if any, are the nonzero solutions of the equation $Av = \lambda v$. We can rewrite this equation as follows:

$$\begin{aligned} & Av = \lambda v \\ \iff \quad & Av - \lambda v = 0 \\ \iff \quad & Av - \lambda I_n v = 0 \\ \iff \quad & (A - \lambda I_n)v = 0. \end{aligned}$$

Therefore, the eigenvectors of A with eigenvalue λ, if any, are the nontrivial solutions of the matrix equation $(A - \lambda I_n)v = 0$, i.e., the nonzero vectors in $\mathrm{Nul}(A - \lambda I_n)$. If this equation has no nontrivial solutions, then λ is not an eigenvector of A.

Eigenspace

Understanding Eigenspace

Finding the eigenvectors for a given eigenvalue means solving a homogeneous system of equations.

Example: If $$A = \begin{pmatrix} 7 & 1 & 3 \\ -3 & 2 & -3 \\ -3 & -2 & -1 \end{pmatrix},$$

then an eigenvector with eigenvalue λ is a nontrivial solution of the matrix equation

$$\begin{pmatrix} 7 & 1 & 3 \\ -3 & 2 & -3 \\ -3 & -2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Eigenspace

Understanding Eigenspace (Continued…)

This translates to the system of equations

$$\begin{cases} 7x + y + 3z = \lambda x \\ -3x + 2y - 3z = \lambda y \\ -3x - 2y - z = \lambda z \end{cases} \longrightarrow \begin{cases} (7-\lambda)x + y + 3z = 0 \\ -3x + (2-\lambda)y - 3z = 0 \\ -3x - 2y + (-1-\lambda)z = 0. \end{cases}$$

This is the same as the homogeneous matrix equation

$$\begin{pmatrix} 7-\lambda & 1 & 3 \\ -3 & 2-\lambda & -3 \\ -3 & -2 & -1-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0,$$

i.e., $(A - \lambda I_3)v = 0$.

Eigenspace

Definition. Let A be an $n \times n$ matrix, and let λ be an eigenvalue of A. The ***λ-eigenspace*** of A is the solution set of $(A - \lambda I_n)v = 0$, i.e., the subspace $\mathrm{Nul}(A - \lambda I_n)$.

Note:

The λ-eigenspace is a subspace because it is the null space of a matrix, namely, the matrix $A - \lambda I_n$. This subspace consists of the zero vector and all eigenvectors of A with eigenvalue λ.

Eigenspace

EXAMPLE ! For each of the numbers $\lambda = -2, 1, 3$, decide if λ is an eigenvalue of the matrix

$$A = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix},$$

and if so, compute a basis for the λ-eigenspace.

SOLUTION

(a) The number 3 is an eigenvalue of A if and only if $\mathrm{Nul}(A - 3I_2)$ is nonzero. Hence, we have to solve the matrix equation $(A - 3I_2)v = 0$. We have

$$A - 3I_2 = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix} - 3\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -4 \\ -1 & -4 \end{pmatrix}.$$

Eigenspace

SOLUTION (Continued...)

The reduced row echelon form of this matrix is

$$\begin{pmatrix} 1 & 4 \\ 0 & 0 \end{pmatrix} \xrightarrow[\text{form}]{\text{parametric}} \begin{cases} x = -4y \\ y = \quad\; y \end{cases} \xrightarrow[\text{vector form}]{\text{parametric}} \begin{pmatrix} x \\ y \end{pmatrix} = y \begin{pmatrix} -4 \\ 1 \end{pmatrix}.$$

Since y is a free variable, the null space of $A - 3I_2$ is nonzero, so 3 is an eigenvector. A basis for the 3-eigenspace is $\left\{ \binom{-4}{1} \right\}$.

Concretely, we have shown that the eigenvectors of A with eigenvalue 3 are exactly the nonzero multiples of $\binom{-4}{1}$. In particular, $\binom{-4}{1}$ is an eigenvector, which we can verify:

$$\begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} -4 \\ 1 \end{pmatrix} = \begin{pmatrix} -12 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} -4 \\ 1 \end{pmatrix}.$$

Eigenspace

SOLUTION (Continued...)

(b) The number 1 is an eigenvalue of A if and only if $\mathrm{Nul}(A - I_2)$ is nonzero. Hence, we have to solve the matrix equation $(A - I_2)v = 0$. We have

$$A - I_2 = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -4 \\ -1 & -2 \end{pmatrix}.$$

This matrix has determinant -6, so it is invertible. By the invertible matrix theorem, we have $\mathrm{Nul}(A - I_2) = \{0\}$, so 1 is not an eigenvalue.

Eigenspace

SOLUTION (Continued...)

(c) The eigenvectors of A with eigenvalue -2, if any, are the nonzero solutions of the matrix equation $(A+2I_2)v = 0$. We have

$$A + 2I_2 = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix} + 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -4 \\ -1 & 1 \end{pmatrix}.$$

The reduced row echelon form of this matrix is

$$\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \xrightarrow[\text{form}]{\text{parametric}} \begin{cases} x = y \\ y = y \end{cases} \xrightarrow[\text{vector form}]{\text{parametric}} \begin{pmatrix} x \\ y \end{pmatrix} = y\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Hence there exist eigenvectors with eigenvalue -2, namely, any nonzero multiple of $\binom{1}{1}$. A basis for the -2-eigenspace is $\left\{\binom{1}{1}\right\}$.

Eigenspace

EXAMPLE 6 Let $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$. An eigenvalue of A is 2. Find a basis for the corresponding eigenspace.

SOLUTION Form

$$A - 2I = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

and row reduce the augmented matrix for $(A - 2I)\mathbf{x} = \mathbf{0}$:

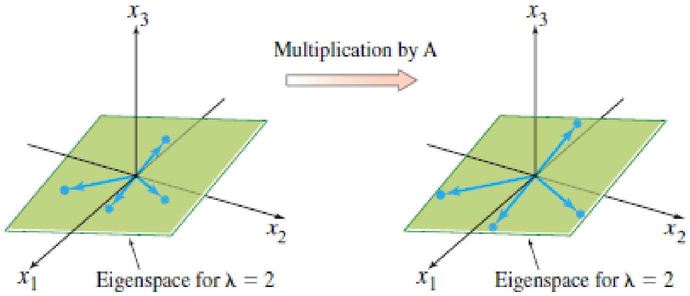
$$\begin{bmatrix} 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Eigenspace

SOLUTION (Continued...)

At this point, it is clear that 2 is indeed an eigenvalue of A because the equation $(A - 2I)\mathbf{x} = \mathbf{0}$ has free variables. The general solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \quad x_2 \text{ and } x_3 \text{ free}$$



The eigenspace, shown in Fig , is a two-dimensional subspace of $\mathbb{R}^3$. A basis is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

■

The Characteristic Equation

Recall

- An ***eigenvalue*** of A is a scalar λ such that the equation $Av = \lambda v$ has a *nontrivial* solution.
- If $Av = \lambda v$ for $v \neq 0$, we say that λ is the ***eigenvalue for*** v
- the nontrivial solutions of the matrix equation $(A - \lambda I_n)v = 0$

$$Av = \lambda v \quad \Longleftrightarrow \quad (A - \lambda I_n)v = 0$$

Useful information about the eigenvalues of a square matrix A is encoded in a special scalar equation called the characteristic equation of A. A simple example will lead to the general case.

The Characteristic Equation

EXAMPLE 1 Find the eigenvalues of $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$.

SOLUTION We must find all scalars λ such that the matrix equation

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

has a nontrivial solution. By the Invertible Matrix Theorem , this problem this problem is equivalent to finding all λ such that the matrix $A - \lambda I$ is *not* invertible, where

$$A - \lambda I = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix}$$

this matrix fails to be invertible precisely when its determinant is zero. So the eigenvalues of A are the solutions of the equation

$$\det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix} = 0$$

The Characteristic Equation

SOLUTION (Continued...)

Recall that

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

So

$$\begin{aligned} \det(A - \lambda I) &= (2 - \lambda)(-6 - \lambda) - (3)(3) \\ &= -12 + 6\lambda - 2\lambda + \lambda^2 - 9 \\ &= \lambda^2 + 4\lambda - 21 \\ &= (\lambda - 3)(\lambda + 7) \end{aligned}$$

If $\det(A - \lambda I) = 0$, then $\lambda = 3$ or $\lambda = -7$. So the eigenvalues of A are 3 and -7. ■